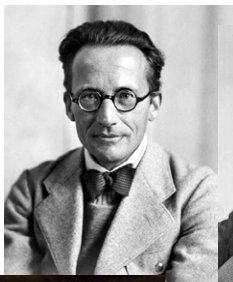


Quantum Atomic Theory

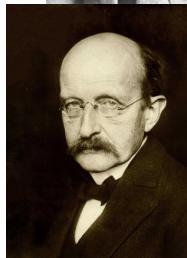
Erwin Schrödinger (1887 - 1961)



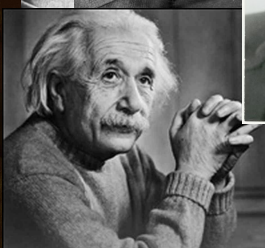
Werner Heisenberg (1901 - 1976)



Wolfgang Ernst Pauli
(1900 – 1958)



Max Plank (1858 - 1947)



Albert Einstein (1879 - 1955)

Bohr-Rutherford vs Quantum

Similarities:

- electrons have discrete amount of energy
- positive nucleus in the centre.

Differences:

Bohr-Rutherford

- atom is 2D
- the location of the electrons can be predicted
- electrons travel on circular paths around the nucleus

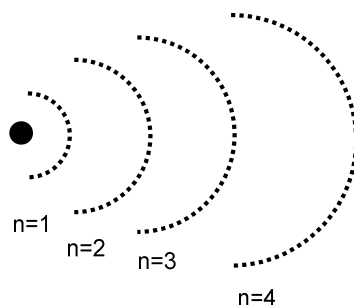
Quantum

- atom is 3D
- the location of the electrons cannot be predicted
- electrons move randomly

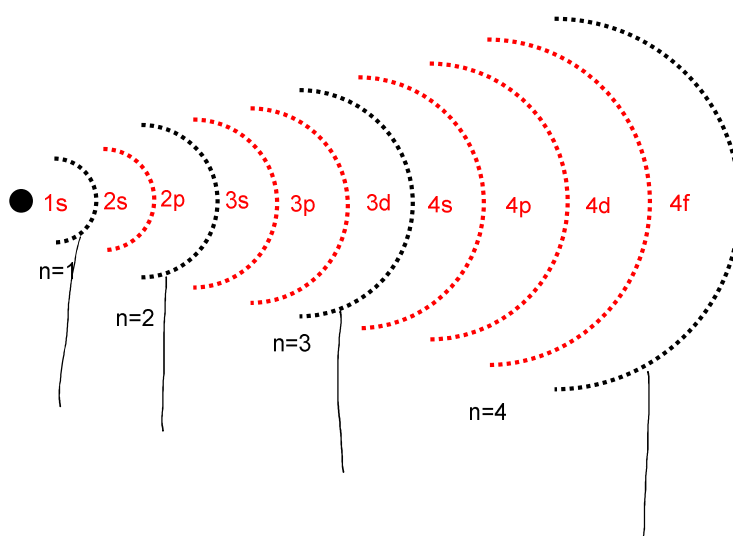
The Heisenberg Uncertainty Principle

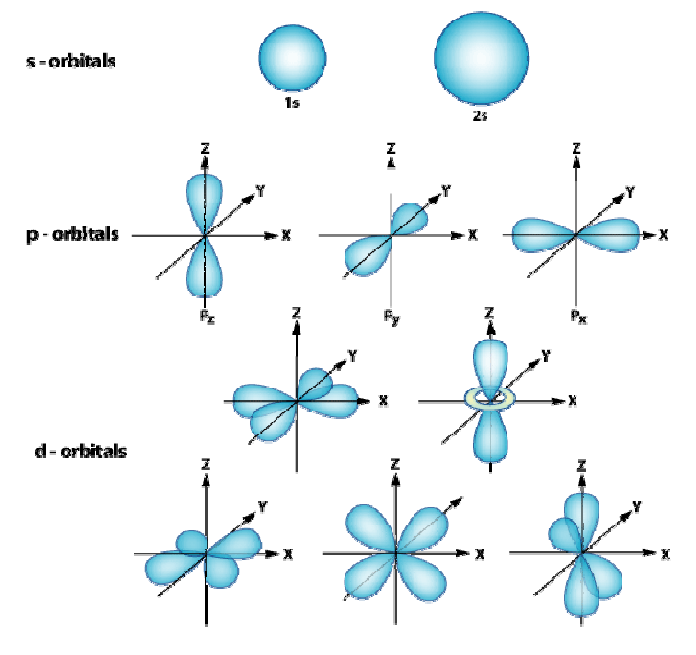
The position and the magnitude of an electron cannot both be known. If one is measured, the other is altered

The Bohr model describes electrons in different orbits around the nucleus



The Quantum model describes electrons in different orbitals (or energy levels) around the nucleus. The traditional orbits of the Bohr model are subdivided.





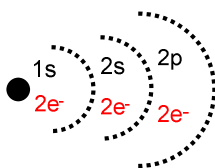
The maximum number of electrons in each type of suborbital

$s = 2e^-$ maximum
 $p = 6e^-$ maximum
 $d = 10e^-$ maximum
 $f = 14e^-$ maximum

Orbitals are filled in the following order:

7s 7p
 6s 6p 6d
 5s 5p 5d 5f
 4s 4p 4d 4f
 3s 3p 3d
 2s 2p
 1s

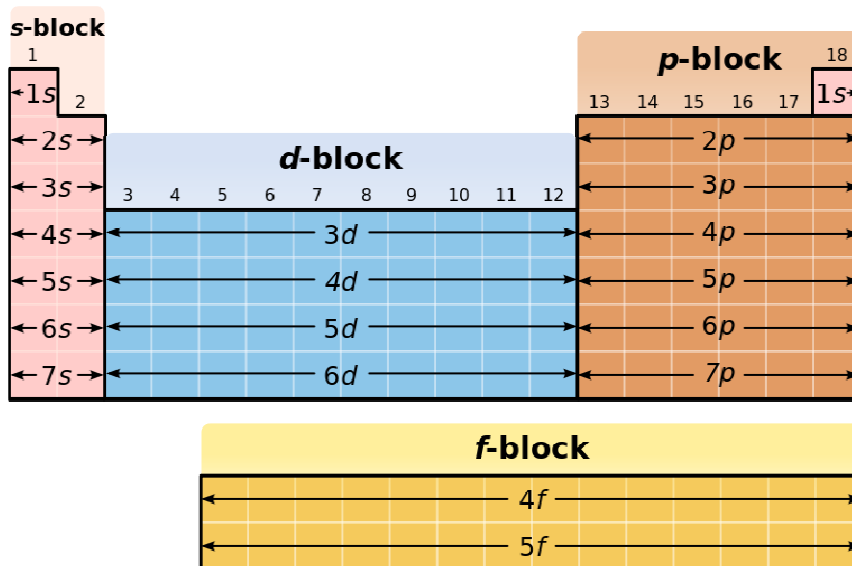
7s 7p
6s 6p 6d
5s 5p 5d 5f
4s 4p 4d 4f
3s 3p 3d
2s 2p
1s



$1s^2 2s^2 2p^2$

Principle #

The superscript refers to the # of electrons in the suborbital

$$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$$


Ex) State the electron configuration of the following:

a) Fe

b) Ni

c) Au

d) Zn^{2+}

7s 7p
6s 6p 6d
5s 5p 5d 5f
4s 4p 4d 4f
3s 3p 3d
2s 2p
1s

Shorthand Electron Configuration

Ne: $1s^2 2s^2 2p^6$

Mg: $1s^2 2s^2 2p^6 3s^2$ or $[\text{Ne}] 3s^2$

noble gas indicates all inner shells are full

Therefore, Au can be shortened from

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^{14} 5d^9$

to

$[\text{Xe}] 6s^2 4f^{14} 5d^9$

Read pg 11-13

State the electron configuration for:

Li, O, P, Mn, Br, & Ag